ChangeLoop

Concept Note

1. Title page

Startup name	ChangeLoop
Tagline	Cut the water by a fifth. The evaporator burns exactly the same fuel.
Date	9 October 2026
Prepared by	Shorya Ashish Mittal, project lead and sole developer
Contact	shoryamittal9653@gmail.com
Working prototype	https://changeloop-water-intelligence.onrender.com

2. Background and problem statement

Context

Tirupur in Tamil Nadu is India's knitwear capital, with over 54% of the country's knitwear exports. In the year to March 2026 it exported around Rs 46,000 crore of knitwear. Its dyeing factories sit on the Noyyal river.

In 2011 the Madras High Court ordered every dyeing and bleaching unit in Tirupur shut to save the Noyyal. Over 700 units and treatment plants closed, 40,000 to 50,000 workers lost their jobs, and the cluster lost an estimated Rs 50 crore a day. The units were allowed back only if they stopped releasing liquid waste completely. The rule is called zero liquid discharge, or ZLD.

Today 360 dyeing units operate in and around the city: 60 with their own treatment plants, and the rest sharing 18 common ones. Between them they treat about 130 million litres of effluent a day and recover about 92% of it for reuse.

It worked for the river. The waste water stopped going in. But it is costly to run. The dyers' association says the 18 common plants spend around Rs 30 crore a month on electricity alone, and the problem moved to the boiler house, where nobody is managing it.

The problem

To recycle the water, a factory has to boil the leftover salty stream until only dry salt is left. Boiling needs heat and the heat comes from steam. Energy audits of more than sixty dyeing units in Tiruppur during 2024 and early 2025 found each one burning on average about 2,000 tonnes of solid fuel a year, primarily wood, with coal and briquettes also in use. Boiling one cubic metre of leftover takes about 188 kWh of heat.

One caveat I want up front rather than buried. The rupee and carbon figures in this note are derived on a coal-fired basis, which is what my coefficient registry states. A wood-fired boiler has different fuel economics and a different carbon profile, so those two conversions need re-deriving for any particular site. The physics underneath does not move: salt sets how much has to be boiled whatever fuel raises the steam.

Here is the part almost nobody acts on.

The amount you have to boil is fixed by the salt, not by the water.

```
leftover volume  =  salt mass  /  strongest the membranes can make it
```

Water goes round the loop and comes back. Salt does not. Salt has to leave as a solid, and the only way out is through the evaporator. So the evaporator must boil whatever amount of water that salt happens to be sitting in.

The result is easy to test, and my system tests it on every request:

Change a factory makes	What happens to evaporator energy
Uses 20% less water	**no change at all**
Puts in 20% less salt	**20% less energy**

Zero, not a small amount. It falls straight out of the arithmetic.

So two changes that look identical on a water meter do completely different things to the evaporator's fuel bill. Most water-saving work in the cluster aims at the water, and leaves the evaporator burning exactly what it burned before.

Why a problem this simple is still not solved

It is fair to ask why, if the arithmetic is this plain, a cluster that exports Rs 46,000 crore a year has not already fixed it. The honest answer is that the saving is not hidden. It is known, documented, and still not taken.

Energy audits of more than sixty dyeing units in Tiruppur during 2024 and early 2025 found that simple efficiency measures could cut about **15% of fuel consumption**. Many of those measures needed little or no investment. Adoption stayed low anyway.

That is the real problem, and it is not a technical one. Four things keep it in place.

An audit produces a report, not a decision. A consultant visits, measures, writes it up and leaves. On Monday the planner schedules the day exactly as before, because nothing in their actual workflow changed. Reports do not run shifts.

Changing anything feels like risking production. The audits name fear of disrupting production as a barrier even for measures that cost nothing. In a job-work cluster running to tight delivery dates, a late or off-shade lot costs far

more than the fuel it might have saved. Any proposal that cannot guarantee the ship date is dead on arrival, however good its numbers are.

The person who decides does not see the bill. The salt load is set by a production planner choosing which lots run in what order and which rinse method to use. They make that call hours before anything shows up. The fuel bill arrives weeks later, in a different department's budget, with no way to trace it back. The planner is not being careless. They are working without the one number that would change their mind.

Nobody meters the thing that matters. Water is metered because it is billed and regulated. Salt is bought by the sack and dissolved into a bath, and no Indian dyehouse I can find reports salt load per lot. What is not measured does not get managed, and here the unmeasured variable is the one driving the cost.

So the gap is not knowledge. It is that no tool puts the consequence in front of the person making the choice, at the moment they make it, without asking them to risk a delivery.

There is a second problem one level up. A factory runs several dyeing machines into one shared evaporator. Each planner picks what is best for their own machine, and every choice is correct on its own. Added together they can still be more than the evaporator can boil. In my model four machines each picking their own best plan put the shared evaporator at **121.9% of what it can handle**. No tool that looks at one machine can see this.

Target audience

The people most affected are the ones running these factories under a rule they did not choose and cannot opt out of.

Who	Why this matters to them
Production planners in Tirupur dyeing units	They make this decision every shift and cannot see what it costs
Factory owners	They pay a fuel bill they cannot trace or control
The 18 shared treatment plants	They buy the steam and store the salt. Over 1,00,000 tonnes of mixed waste salt now sits in sheds on their premises, with no disposal method yet found
Households and farmers in the Noyyal basin	Groundwater here is heavily stressed. Every litre not pulled out matters more than a litre saved somewhere easy

3. Vision and mission

Vision

Every closed water loop in Indian industry is run on what it actually costs, not on what is easy to measure.

Water meters are everywhere and they measure the wrong thing. In a closed loop the environmental and financial damage is driven by dissolved load, and almost nothing in industry is set up to see that. The long-term aim is to make the hidden number visible wherever this physics applies, starting with textiles and extending to cooling towers, boiler blowdown and other closed systems.

Mission

Put the downstream cost in front of the person making the upstream decision, at the moment they make it, with every number open to inspection.

In practice that means three things. Build the decision layer and prove the physics. Prove it on one factory with real instruments. Then reach the rest of the cluster through the shared treatment plants, whose own economics already point the same way.

4. Solution overview

What it does, in one paragraph

ChangeLoop sits where the schedule is made. Before a shift starts, it takes the lots that have to run and works out every order they could run in, and every rinse method available for each. For each of those combinations it computes the full consequence: water drawn, salt added, leftover volume to boil, evaporator steam, CO2 and rupees, and whether any delivery date would be missed. It throws away everything that ships late, ranks what is left, and shows the planner the best plan with its working.

The planner can accept it or ignore it. Nothing else changes: no new equipment, no new chemistry, no change to who approves what.

Why this is the right place to intervene

Every other way of attacking this problem asks the factory to change something physical, and that is why they stall.

Buy a better evaporator and you have spent capital on treating the symptom. Buy low-salt chemistry and you have raised the cost per kilogram today for a fuel saving you cannot yet see. Run an energy audit and you get a report. All three ask for money or risk up front against a benefit nobody has measured.

Changing the order lots run in costs nothing and is reversible the same day, and the tool never offers an order that misses a ship date. It is the one lever in this process

that is free to pull. What I have added is making it a visible lever rather than an invisible one.

What makes it different

One line, if you only read one. Every planning tool in a dyehouse optimises for throughput and delivery. ChangeLoop is the only one I have found that prices what the schedule does to the boiler, and it does that while the schedule is still being chosen.

The rest follows from treating that as a safety-critical decision rather than a dashboard.

	How ChangeLoop behaves	Why it matters here
Delivery dates	A plan that ships late is deleted from the search, not scored badly	Removes the single objection that kills every efficiency proposal in a job-work cluster
Every input number	All 25 are labelled published, derived, assumed or measured, with the source and the arithmetic, readable in the live system	A process engineer can audit the model instead of trusting it. None is MEASURED yet and a test enforces that
Authority	Issues no machine settings and cannot release a dye bath. Early release needs a named person	The quality supervisor keeps the decision that carries the quality risk
Bad sensor data	Holds and claims zero saving rather than estimating	An advisory tool that quietly guesses is worse than no tool
Credit for savings	A recommendation the planner declines counts as zero, even though the engine computed it	The ledger records what was done, not what was advised, so it can be billed against
More than one machine	Compares what each machine would pick alone against the cheapest combination the shared evaporator can take	Four locally correct choices can still breach the plant, and no single-machine tool can see it

Value proposition

Who	What they get
Factory owner	Lower fuel and water bills from a scheduling change, with nothing to buy and no change to the approval process
Production planner	The missing number on their screen, at the moment they need it, with the ship date protected by construction
Shared treatment plant	Less salt arriving, so less steam bought and less solid salt stored

Who	What they get
Regulator or tariff-setter	A figure for what a price change would actually do, instead of an argument in principle

5. Objectives and goals

Short term, the next 12 months

Objective	Measure of success
Install instruments at one pilot factory	Steam flow and reject conductivity metered for 28 days at over 95% completeness
Prove the recommendations are sound	Planner agrees with over 60% of them unprompted, during four weeks where the tool changes nothing
Measure a real saving	Water and steam reduction measured with a confidence interval, on alternate weeks on and off
Keep quality intact	Zero colour-fastness failures attributable to an approved plan
Turn assumptions into measurements	Move the two heaviest input numbers from published to measured

Long term, two to five years

The first two are the ones that matter. They change the cluster whether or not anyone buys the software.

Objective	Target	Why it has leverage
Get treatment plants charging by salt, not by volume	Two CETPs running a salt-indexed tariff for their members	As far as I can find, CETP charges are set by the volume a member sends, not by how salty it is, so the cost of the salt a unit creates lands on the plant. Change that and every member has the incentive on day one. This is the highest-leverage outcome in the project, and since section 6 it is also what I sell rather than something I hope somebody does
Make salt load per lot a number the industry reports	Salt per lot metered and reported at 40 units, the way water already is	Water got managed once it got metered. Nothing gets managed until it is measured, and right now nobody measures the variable that drives the cost

Objective	Target	Why it has leverage
Close the dye premium gap by buying together	A joint procurement approach across CETP members, aimed at the 12% gap my engine computes	12% is small enough that aggregated demand across a few hundred units could close it. Then the 47% plan becomes the cheap plan and nobody has to be persuaded
Solve cluster coordination properly	A method that works for hundreds of member units, not the 256 combinations the current solver checks exhaustively	The current approach is proven but will not scale, and I would rather say that than imply it will
Put the evidence in the open	The measured coefficients published, with the method, for anyone to use or contest	A model nobody can check is worth less than one people argue with
Extend beyond textiles	One validated application in another closed-loop system, such as evaporative cooling	The physics is not specific to dyeing, but it is unproven anywhere else

6. Business model

Who actually pays for salt

My first business model billed the factory a share of its savings, and the arithmetic talked me out of it. A dyeing unit buys salt at about Rs 9 a kilogram and buys water by volume. Salt is cheap to a factory. That is precisely why nobody acts on it, and charging the party with the weakest incentive is a hard way to build a business.

Then I worked out who does carry the cost.

Every tonne of salt that reaches a treatment plant forces about 17 cubic metres of reject to be boiled. Using my own coefficients, that is Rs 6,363 of steam, Rs 3,000 of evaporator operating cost and Rs 527 of power: **Rs 9,891 a tonne**.

A single CETP in Tirupur handles roughly a sixth of the cluster's 100 million litres a day. At the recovery rates these plants report, that is about **7,700 tonnes of salt a year, costing it around Rs 7.62 crore** to deal with.

It chose none of it. Its seventeen member factories did. And because the plant bills members by the volume they send rather than how salty it is, it has no way to charge for any of it. The dyers' association says the 18 plants together spend around Rs 30 crore a month on electricity, and this is a large part of why.

That is the same misaligned incentive as section 2, one level up. The factory decides, somebody else pays, and nobody can see the link.

How it makes money

So I sell to the plant, and give the planner to the factory cheaply enough that adopting it is not a decision anyone has to defend.

Who pays	What for	Price
Treatment plant	Salt attribution: how much salt each member sent, what it cost the plant, and a tariff engine to bill for it	Rs 12,00,000 a year, plus 20% of the verified reduction
Treatment plant, once	Onboarding: conductivity meters on member discharge lines, attribution set up against the plant's own costs	Rs 2,50,000
Member factory	The planner itself, priced per lot so it scales with use	Rs 15 a lot, about Rs 75,000 a year

A 3% cut in salt arriving saves a plant about Rs 22,80,000 a year, so the platform pays for itself in the first year at a reduction small enough to be plausible. At 10% it saves about Rs 76,20,000.

Why this way round

The factory ask becomes signable. Rs 75,000 a year is an operating expense a dyeing-unit owner approves without a board meeting. A share of savings on my earlier model would have been Rs 19,65,000, and no owner signs that for software against a saving nobody has measured yet.

Eighteen conversations instead of three hundred and sixty. One plant relationship reaches about seventeen factories. That is the difference between a business one person can start and one that needs a sales team on day one.

It creates pull instead of push. Once a plant bills by salt load, its members suddenly need a way to manage salt, and the planner is that way. I stop having to persuade anyone that the problem is real; their invoice does it.

It is what the system is uniquely able to do. Plenty of software can schedule a dyehouse. Attributing a plant's salt load back to the member and the lot that caused it is the thing only this engine does, because it is the thing the whole model was built to compute.

I am paid on what was measured and approved. Savings are computed on the server, counted only when a named person approved the plan, and written to a tamper-evident ledger. That ledger is the billing basis, which turns the project's weakest point into the mechanism: nothing is measured yet, so nothing is owed yet.

What a factory actually gets

The engine reports a net saving per lot against a conventional single-stage rinse. To turn that into a year I need to know how many lots a real unit runs, and I would rather anchor that on something measured than multiply my own model out.

The audits put a Tiruppur dyeing unit at about 2,000 tonnes of fuel a year. Seasoned wood gives about 15.5 MJ per kilogram, a boiler delivers about 78% of that as heat, and my engine uses about 1,346 kWh of energy per lot. Put together, that fuel supports roughly **5,000 lots a year**, and nearer 4,000 if the wood is wet. I use 5,000.

	Plan it recommends today	Low-salt plan
Net saving per lot	Rs 1,572	Rs 783
Lots a year, mid-size unit	5,000	5,000
Net saving a year	**Rs 78,60,000**	**Rs 39,14,000**
My fee at Rs 15 a lot	Rs 75,000	Rs 75,000
Factory keeps	**Rs 77,85,000**	**Rs 38,39,000**
Water and steam cut	6%	47%

Read that table carefully, because it is the point of the whole project.

The fee is the same in both columns because it is charged per lot, not per rupee saved. The factory keeps essentially all of the benefit.

The plan the optimiser recommends today saves **more money** but only 6% of the fresh water. The low-salt plan saves **less money** and nearly half the water and steam. The dye premium eats most of the saving, which is exactly why a profit-seeking factory does not buy it and why the cluster keeps burning fuel it does not need to.

Close the 12% gap in the dye premium and the low-salt plan becomes the cheaper one too. Then the factory saves more money and more water at the same time, and nobody has to be persuaded to choose between them.

Where the saving actually comes from

The recommended plan does two things at once: it re-orders the lots, and it switches the rinse to a counter-current cascade. They are worth separating, because they behave very differently.

Source of the saving	Per lot	Per year, 5,000 lots	Effect on fresh water and evaporator steam	Who gets it
Re-ordering the lots	Rs 607	Rs 30,37,000	All of the 6% cut	Every unit, whatever rinse it already uses
Switching to counter-current rinsing	Rs 964	Rs 48,22,000	None at all	Only units still on a single-stage rinse
Both together, the plan it recommends	**Rs 1,572**	**Rs 78,60,000**	**6%**	

Rows may not add exactly because of rounding.

The second row is worth a moment. Counter-current rinsing cuts the water circulated through the machines by about a quarter, and changes the fresh water drawn and the evaporator steam by exactly nothing, because it does not change the salt. It is the same finding as section 2, turning up on its own inside the engine.

It also answers the obvious objection. Counter-current cascade is an established best available technique, and a unit already running it has already taken the rinse part of the saving. It still gets the re-ordering part, about Rs 607 a lot, and all of the cut in fresh water and evaporator steam, because that comes from the order, not the rinse. The first thing a pilot does is establish which kind of unit it is in.

All of these rupees are modelled, on a modelled order book of five lots. The direction and the ranking are what I would defend; a pilot establishes the amounts.

How customers are reached

Through the shared treatment plants, which are the customer. Tirupur has 18 of them serving about 300 units. One conversation reaches about seventeen factories instead of one, which is the difference between a sales effort one person can run and one that needs a team from the start.

They are not only a channel. The Rs 7.62 crore a year of salt-driven cost is theirs, so the attribution platform is bought for their own reasons, not as a favour to their members.

Through running silently first. Four weeks of being right while changing nothing is how a tool earns the right to influence a factory's decisions. It costs the factory nothing.

Through the dyers' own association. The Dyers Association of Tiruppur already speaks for its members on salt disposal and effluent, which is exactly the problem

this addresses.

Partnerships that would help

Partner	What it brings
A shared treatment plant in Tirupur	The pilot site, the steam data, and the route to its member units
A dye chemistry supplier	Real pricing on low-salt chemistry, which is the lever with the smallest gap to close
A state pollution control board or CPCB	Credibility, and access to the measured data that would replace my published figures
An academic water or chemical engineering group	Independent review of the model before anyone relies on it

7. Market analysis and competition

Market size

Level	Size	Basis
Tirupur, immediate	18 treatment plants at about Rs 20,00,000 a year is Rs 3.6 crore, and 360 factories at Rs 75,000 is Rs 2.7 crore. Together of the order of **Rs 6.3 crore a year**	360 units operating in 2025, 300 of them on 18 CETPs, per the Dyers Association of Tiruppur. The plant fee is anchored on a cost I can derive, not on a saving nobody has measured, which is why this is far below what a share-of-savings model would have claimed
Indian textile ZLD, wider	Tirupur is one of several clusters under ZLD rules. Surat, Erode, Karur, Ludhiana, Panipat and Jetpur have similar processing bases. The whole Indian ZLD equipment market across all industries is put at about **USD 1.33 billion in 2025**, growing to USD 2.52 billion by 2032	The dollar figure is a commercial market estimate covering all ZLD, not textile software. I have not counted dyeing units outside Tirupur and will not present a number I cannot source
Adjacent closed loops, later	Evaporative cooling, boiler blowdown and mine water share the same physics	Unvalidated. Listed as direction, not as market

The growth driver is regulation that already exists. ZLD is compulsory and already paid for. ChangeLoop does not ask anyone to install new equipment, it makes equipment they already own cheaper to run.

The second driver is carbon reporting. European buyers increasingly ask Indian suppliers for emissions data. The EU carbon border mechanism does not cover textiles today, since it applies to iron and steel, cement, aluminium, fertilisers, electricity and hydrogen, but textiles have been named as a candidate for a later phase. If that happens, a tool that can show measured and traceable reductions in process emissions becomes a compliance asset as well as a cost saver. I am treating that as a possible tailwind, not as a commitment anyone has made.

Competitive landscape

Who	What they do	Why ChangeLoop is different
Dyehouse software, such as SedoMaster and Datatex	Plan production, manage recipes, track batches	They schedule for throughput and delivery. Neither prices the downstream thermal cost of the schedule, which is the whole point here
ZLD equipment companies, such as Praj and VA Tech Wabag	Build and run the membranes and evaporators. Praj has worked on Tirupur textile effluent; VA Tech Wabag builds evaporator-based ZLD plants	They sell the plant. They do not touch the upstream decision that sets how hard the plant has to work
Water monitoring and IoT dashboards	Report flows and quality after the fact	They measure water volume, which the table in section 2 shows is the wrong number for the evaporator's fuel bill
Cleaner dye chemistry, such as Archroma's Avitera reactive range, and CIRCOT's low-salt dyeing technology in India	Sell or license chemistry that cuts salt, water or energy in the dye bath	They sell the lever. They cannot tell a factory when it pays, which is exactly what my system computes
Consultancies and energy audits	One-off studies	A report is read once. This runs every shift and is checked against what was actually approved

The honest summary: the pieces exist. The membranes exist, the chemistry exists, the planning software exists. What I have not found anywhere is something that connects the upstream scheduling decision to the downstream fuel bill, which is where the saving is hiding.

8. Implementation plan

Current status

The prototype is built and running. It is a live system anyone can open, not a mockup.

API	34 endpoints. Every saving is worked out on the server, so the browser cannot claim one
Tests	190, covering the physics, the safety rules, concurrency and the interface
Input numbers	25, each labelled with its source. 17 published, 4 derived, 4 assumed, 0 measured
Validation	Fed the water strength CPCB measured at a Tirupur unit, 18,340 mg/L, the model predicted 30.6% leftover. Indian plants report 20 to 30%

Timeline

Stage	Weeks	What happens	Gate to pass
1. Measure the baseline	1-4	Instruments installed. Nothing changes on the floor	28 days of data at over 95% completeness
2. Run it silently	5-8	The tool recommends, nobody acts	Planner agrees with over 60% unprompted
3. Planner may approve	9-16	Reordering only, no chemistry change	Measured water reduction, zero off-shade lots
4. Early release trial	17-24	Quality supervisor may allow early wash-off release	Every released lot lab tested. One failure ends the trial
5. Decide on chemistry	25-32	With steam cost now measured, decide if low-salt dye pays	A decision backed by measured, not assumed, numbers
6. More factories	33-52	Second and third unit on the same treatment plant	Two more units live and paying

The weeks run on and off in alternate blocks, so the comparison is against the same season, the same order mix and the same crew. The whole point is to answer one question: did the tool cause the saving, or would it have happened anyway?

Key milestones

Milestone	When	Why this date
First pilot site agreed	Month 4	Three months of introductions, site visits and a safety review before anyone lets a student near a steam header

Milestone	When	Why this date
First measured coefficient replacing a published one	Month 6	Instruments in, 28 days of clean data
First saving measured with a confidence interval	Month 10	Needs alternate on and off weeks through stage 3
First treatment plant signs for attribution	Month 15	After the pilot has a measured result to show
Salt-indexed tariff piloted at one plant	Month 24	The plant's members have to agree to it, which is a governance step, not a technical one
3 plants and 30 factories live	Month 36	About the ceiling for one person without hiring

9. Financial overview

All figures in this section are plans, not results. I have no revenue and no customers today. The software already exists and runs, so what follows is the cost of proving it on a real factory, not the cost of building it.

What the pilot actually costs

Instrumentation per site. These are the two readings that turn the heaviest assumptions in the model into measurements.

Item	Cost	Why this figure
Vortex steam flow meter, DN50, steam rated	Rs 60,000	Indian suppliers quote Rs 35,500 to Rs 96,000 depending on line size and specification
Inductive conductivity transmitter and sensor	Rs 50,000	The reject stream fouls contacting electrodes, so an inductive probe is the correct type. Indian industrial units run Rs 25,000 to Rs 60,000
Tapping, wiring and commissioning	Rs 35,000	Hot line tapping and a safe install on an existing steam header
Data logger and gateway	Rs 15,000	So readings reach the system without someone copying them down
Calibration and first-year checks	Rs 15,000	A drifting sensor is worse than no sensor
Per site	**Rs 1,75,000**	

The whole pilot across twelve months, for two sites.

Item	Cost	Why this figure
Instrumentation, two sites	Rs 3,50,000	Two sites so the result is not one factory's quirk
Independent colour-fastness testing	Rs 50,000	About 150 samples. SITRA in Coimbatore publishes Rs 300 to Rs 350 per sample for colour fastness to washing
Travel to Tirupur, about twelve visits	Rs 60,000	About Rs 5,000 a visit including travel and a night's stay, assuming travel within Tamil Nadu. From further away it is closer to double. Install, calibrate, and sit with the planner during the shadow weeks
Cloud hosting and domain	Rs 20,000	The system runs on a small instance today
Contingency, about 10%	Rs 48,000	
Total	**Rs 5,28,000**	

What costs nothing

The software is built, tested and running. Extending it to a second or third factory is configuration, not development.

The host factory contributes in kind rather than in cash: access to the steam header and the reject line, an extract from its planning system, and some of the planner's time during the weeks when the tool changes nothing. None of that is a cheque, and without it the cash above buys nothing.

My own work on the pilot is not costed here. I am doing it either way.

How a pilot like this gets funded

Three routes, and they are not exclusive. A treatment plant can co-fund the instruments, because the steam saving lands on its own bill. A state or central scheme for cluster-level environmental work can cover metering. Or the first site pays a reduced fee and keeps the saving, which is the cleanest test of whether anyone actually wants this.

I am not putting a funding request in this note. The number above is here so the cost of the next step is visible, and because a plan that cannot say what it costs is not a plan.

Unit economics

Item	Amount
Treatment plant platform	Rs 12,00,000 a year
Treatment plant onboarding, once	Rs 2,50,000
Share of the plant's verified reduction	20%
Member factory planner	Rs 15 a lot, about Rs 75,000 a year

The plant fee is anchored on a cost I can derive rather than a saving nobody has measured. A plant carries about Rs 7.62 crore a year of salt-driven cost, so Rs 12,00,000 is about 1.6% of it, and a 3% reduction returns about Rs 22,80,000. That is a case a plant manager can check without trusting my model.

The factory fee deliberately does not scale with savings. At Rs 15 a lot it is roughly 1% of what the engine says the plan saves, which means I am never arguing with an owner about whether the saving was real.

Revenue projections

The binding constraint on these is not demand. It is that I am one person.

	Year 1	Year 2	Year 3
Treatment plants live	0, pilot only	1	3
Member factories live	2, pilot	10	30
Plant platform fees	none	Rs 12,00,000	Rs 36,00,000
Plant onboarding, new plants only	none	Rs 2,50,000	Rs 5,00,000
Factory planner fees	waived during pilot	Rs 7,50,000	Rs 22,50,000
Total revenue	**nil**	**Rs 22,00,000**	**Rs 63,50,000**

Year 1 earns nothing on purpose. The two pilot factories pay nothing while their baseline is being established, because the whole point of that year is to find out whether the saving is real, and charging for it would bias the answer.

Year 2 assumes one plant signs after seeing pilot results, bringing about ten of its members. Year 3 assumes two more plants. Onboarding is billed once per plant, so year 3 bills two, not three.

The 20% share of verified reduction is deliberately left out of these totals. I can compute what it would be, but it rests on a reduction no instrument has confirmed, and a projection that depends on my own model being right is not a projection.

Three plants and thirty factories by year three is roughly the ceiling for one person doing integration, training and support. Past that I would need to hire, and I would rather show a number I could deliver alone than one that quietly assumes a team I do not have.

10. Impact assessment

If one factory adopts it

For the modelled five-lot queue, on the plan the tool recommends under today's prices, against a conventional single-stage rinse:

	Fresh water	Evaporator steam	Cost
Baseline	11,907 L	2,239 kWh	Rs 33,897
Recommended today	11,187 L (-6%)	2,104 kWh (-6%)	Rs 26,038
Available with low-salt chemistry	6,369 L (-47%)	1,198 kWh (-47%)	Rs 29,984

If the cluster adopts it

A straight-line projection across 400 units at 900 lots each a year. The 400 is the figure my engine uses, a little above the 360 units operating today. It is still deliberately cautious overall: it assumes 900 lots a unit a year against the 5,000 used in section 6, so its totals sit well below what a single unit's own figures would multiply out to. This is a projection and my system labels it as one. It assumes every factory resembles the modelled one and that every recommendation is approved. Neither has been tested, and it must not be read as achieved impact.

Saved per year across the cluster	
Fresh water	75 million litres
Salt	4,500 tonnes
Evaporator steam	14,106 MWh
CO2	6,376 tonnes

Environmental benefit

Less groundwater pulled out of a basin that is already stressed. Less solid fuel burned for the same output, which in Tiruppur mostly means less wood. Less solid salt added to the more than 1,00,000 tonnes already stored in sheds at Tirupur's treatment plants, for which no disposal method has yet been found.

Economic benefit

A lower cost base for a cluster that exports around Rs 46,000 crore a year and competes on price. The 2011 shutdown is living memory in Tirupur, so anything that strengthens environmental compliance also protects the jobs that depend on staying open.

The policy finding

My system can compute what price would make the bigger saving worth buying. It walks each price up or down, re-running the whole calculation at every step, and finds the point where a factory chasing profit picks the 47% plan on its own.

Price	Today	Changes at	Move needed
Low-salt dye premium	Rs 6.50/kg fabric	Rs 5.74/kg fabric	**12% cheaper**
Boiler steam	Rs 2.03/kWh heat	Rs 3.13/kWh heat	54% dearer
Fresh water	Rs 45/m3	Rs 251/m3	457% dearer

The steam row converts into a carbon price. Steam would need to rise Rs 1.095 per kWh of heat, and making that heat gives off 0.452 kg of CO2, so the carbon price that produces the rise is **Rs 2,423 a tonne, about 21 euro**.

The EU carbon market was charging 82.40 euro a tonne on 5 October 2026, close to four times more. So this saving is not waiting for an impossible price. It is waiting for any price at all.

Three caveats travel with that figure. It assumes the whole carbon cost reaches the factory as a higher steam price, which is the best case. It is the point where this decision flips in my model, not a recommendation for what a carbon price ought to be. And it is computed on a coal-fired boiler, so at a wood-fired site the carbon lever is weaker and the dye premium matters more.

The direction does not depend on the fuel, but the size of the gap does. Every switching price is computed against my coal-based steam cost. Where steam is cheaper, saved steam is worth less and the gap is wider: if steam cost a quarter less, the dye premium would need to fall about 17% instead of 12%. The dye premium itself is one of my four assumed coefficients, so the 12% is a computed result on an assumed input, and the first thing a chemistry supplier conversation would do is replace it with a real quote.

11. Conclusion

ZLD saved the Noyyal river and left the cluster with a fuel bill nobody is managing. That bill is set by salt, not water, so most of the water-saving work in Tirupur is aimed at the wrong number. ChangeLoop shows the planner what their choice actually costs, at the moment they make it.

The prototype is built, running and open to inspection. The physics is checked against published plant measurements. The business model follows the money rather than the obvious buyer: a treatment plant carries about Rs 7.62 crore a year of cost created by salt it never chose, so it buys the attribution that lets it charge for it, and its member factories get the planner for about Rs 75,000 a year. Eighteen plants, not three hundred and sixty factories, is a cluster one person can actually

reach.

What it does not have is a single measured number from a real factory, and I would rather say that plainly than let a reviewer discover it. Closing that gap costs about Rs 5,28,000 and needs one factory willing to let me put two instruments on its pipework.

What would help most

None of these is money.

- **An introduction to a shared treatment plant** in Tirupur willing to host a pilot. This is the single thing standing between the project and its first measured number.
- **A technical review** of the model by someone who knows textile effluent. I would rather be told a coefficient is wrong now than after a factory has relied on it.
- **An introduction to a dye chemistry supplier**, to put a real price on the lever my system says has the smallest gap to close.

Next steps

I can demonstrate the live system in under ten minutes, including the parts that show its own limits. I would welcome a technical review of the coefficients before any pilot begins, and I am happy to be told a number is wrong. Every one of them is open at the link below.

https://changeloop-water-intelligence.onrender.com

12. Appendix

How the figures in this note can be checked

Every engine figure quoted here is produced by the live system and can be reproduced. The input numbers, with their sources and arithmetic, are at https://changeloop-water-intelligence.onrender.com/api/evidence

A script in the repository re-checks every number in this document against the engine and fails if any has drifted.

Evidence position

Label	How many	What it means
Published	17	From CPCB, the Central Electricity Authority, Tamil Nadu tariff orders, state pollution board data, the EU reference document, steam tables
Derived	4	Worked out from those by arithmetic a test re-checks

Label	How many	What it means
Assumed	4	My assumptions, labelled as such. All four are commercial prices public sources do not settle
Measured	0	No instrument on a real factory has fed this system. A test enforces this

What this project has not done

- No deployment, no customer, no pilot, no measured result.
- The baseline is a conventional single-stage rinse. Units already on counter-current cascade have taken the rinse part of the saving, about Rs 964 a lot, and keep the re-ordering part. I do not know what share of Tirupur they are.
- The 12% dye premium gap is computed on an assumed premium of Rs 6.50 a kilogram of fabric. It is the coefficient I most want replaced by a quote.
- The cluster figure is a projection, not achieved impact.
- The financial projections in section 9 are plans with stated assumptions. The pilot costs are built from quoted supplier and lab prices. The revenue figures rest on a modelled per-lot saving that no instrument has confirmed, and I have priced the projection at about half of the re-ordering share for that reason. The 5,000 lots a year is derived from audited fuel use, not measured at a site.
- The shared-evaporator solver checks every combination, which will not scale to a treatment plant with hundreds of members.
- The 60,000 mg/L limit is one modelled figure inside a published range of 15,000 to 80,000. Whether it holds in real running conditions is unknown.
- The rupee and carbon conversions assume a coal-fired boiler. Tiruppur units mostly burn wood, so both need re-deriving per site. The physics does not depend on the fuel; the size of every switching price does.
- The data centre and cooling tower idea is an argument from structure, not a result.
- There is no trained model in this system and I do not call it AI-powered. The decision record is a hash chain, not a blockchain.

Sources

Figure	Where it comes from
Tirupur exports of about Rs 46,000 crore in FY26 (April 2025 to March 2026)	Tiruppur Exporters Association, as reported in trade press

Figure	Where it comes from
Over 54% of India's knitwear exports; 2011 closure of over 700 units and plants, 40,000 to 50,000 jobs, about Rs 50 crore a day; 360 units, 60 with their own plants, 18 CETPs; 130 million litres a day treated, 92% recovered; over 1,00,000 tonnes of mixed waste salt in sheds; around Rs 30 crore a month on CETP electricity	Economic Times, 25 June 2025, quoting the Dyers Association of Tiruppur
Salt backlog of about 60,000 tonnes in 2022, for comparison	Dyers Association of Tiruppur, reported by Ecotextile News, October 2022
EU carbon border mechanism covers iron and steel, cement, aluminium, fertilisers, electricity and hydrogen, not textiles	European Commission scope; textiles named as a candidate for a later phase
Indian ZLD market of about USD 1.33 billion in 2025, USD 2.52 billion by 2032	MarkNtel Advisors, India Zero Liquid Discharge Market report. A commercial estimate, cited as such
Archroma Avitera SE reactive dye range, reported to cut water and energy in dyeing by about half	Ecotextile News, January 2024
Praj on Tirupur textile effluent; VA Tech Wabag on evaporator-based ZLD	Company and trade press reporting
Tiruppur dyeing units burn about 2,000 tonnes of solid fuel a year each, primarily wood	Energy audits of more than sixty dyeing MSMEs in Tiruppur, 2024 to early 2025, reported by India Development Review
About 15% of fuel consumption available from simple efficiency measures, with adoption still low	The same energy audits of more than sixty Tiruppur dyeing MSMEs, 2024 to early 2025
Water strength of 18,340 mg/L, leftover at 20-30%, evaporator steam use	Central Pollution Control Board assessment of textile dyeing units and ZLD at Tirupur
Grid emissions, 0.705 kg CO2 per kWh	CO2 Baseline Database for the Indian Power Sector, Central Electricity Authority
Carbon price of 82.40 euro a tonne, 5 October 2026	European carbon prices, S&P Global Commodity Insights
Fresh water at Rs 45 a kilolitre	Reported price of Bhavani river water supplied to Tirupur dyeing units
CETP treatment charge of Rs 185 a kilolitre	Midpoint of the Rs 150 to 220 charged by Tirupur CETPs, Tamil Nadu pollution board plant-level data
Steam at Rs 2.03 a kWh of heat	Derived from published imported coal pricing at 78% boiler efficiency

Figure	Where it comes from
Electricity tariff	Tamil Nadu Electricity Regulatory Commission tariff order, March 2025
Low-salt dye premium of Rs 6.50 a kg of fabric	My assumption, sized against dye cost, labelled ASSUMED in the registry
Seasoned wood at about 15.5 MJ a kilogram	FAO, comparison of energy values for fuelwood at 20% moisture
Latent heat, 0.62694 kWh/kg	Steam tables: 2,257 kJ/kg at 100 C and one atmosphere, divided by 3,600
Evaporator energy band, 150-250 kWh/m3	EU Best Available Techniques reference document for textiles. Used as a check, never as an input